Superstore Data Analysis Project

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Course	Data Analysis
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Tool Used	SQL Microsoft Server, Power BI, Excel, Claud

1. Project Overview

This project demonstrates an end-to-end data analysis workflow using the Sample Superstore dataset. The work covers relational database design in SQL Server, ETL and data cleaning, structured SQL querying, and an interactive Power BI dashboard that surfaces actionable business insights.

1.1 Objectives

- Design a normalised relational database with proper PK/FK constraints
- Import and clean real-world transactional data from a flat CSV source
- Write meaningful SQL queries covering joins, aggregations, and subqueries
- Build an interactive Power BI dashboard with DAX measures and slicers

1.2 Dataset Description

The Superstore dataset contains approximately 9,994 order rows spanning 2014–2017, covering U.S. retail transactions across customer segments, product categories, and shipping modes.

Attribute	Detail
Source	Sample Superstore (CSV flat file)
Records	~9,994 order rows
Time span	2014–2017
Geography	United States — 49 states
Categories	Furniture, Technology, Office Supplies
Customer segments	Consumer, Corporate, Home Office

2. Database Design

The flat CSV was normalised into four tables following third normal form (3NF), eliminating repeating groups and transitive dependencies. The design follows a star-schema pattern with Orders as the central fact table.

2.1 Table Structure

Table	Primary Key	Foreign Keys	Columns	Role
Customers	CustomerID	—	3	Customer dimension
Products	ProductID	—	4	Product dimension
Geography	PostalCode	—	5	Location dimension
Orders	RowID	CustomerID, ProductID, PostalCode	12	Fact table

2.2 Design Decisions

- Used MAX() aggregation with GROUP BY to resolve duplicates in the flat source before inserting into dimension tables.
- Converted Profit field from FLOAT to DECIMAL(18,4) to prevent floating-point rounding errors.
- All foreign key constraints are enforced at the database level to guarantee referential integrity.
- A Calendar dimension table was created in Power BI using DAX to enable SAMEPERIODLASTYEAR time-intelligence.

3. SQL Queries

Six analytical queries were written against the normalised schema, each addressing a specific business question.

#	Business Question	SQL Features Used
1	Top 10 products by total sales	JOIN, GROUP BY, ORDER BY, TOP
2	Orders shipped via First Class	WHERE filter
3	Total profit per state (ranked)	JOIN, GROUP BY, SUM, ORDER BY
4	Products with net losses	JOIN, GROUP BY, HAVING, SUM
5	Most demanded category by quantity	JOIN, GROUP BY, SUM
6	Full orders fact table inspection	SELECT *

3.1 Query Highlights

Query 3 — Profit by State: Joins Orders and Geography, groups by State, and ranks descending by total profit. New York and California lead consistently.

Query 4 — Loss-Making Products: Uses a HAVING clause to surface product groups where SUM(Profit) < 0, identifying Furniture sub-categories as key loss drivers.

Query 5 — Category Demand: Aggregates Quantity by Category. Office Supplies dominates volume while Technology generates higher revenue per unit.

4. Power BI Dashboard

The dashboard connects directly to SQL Server (imported mode). A Calendar table generated in DAX enables time-intelligence measures. All visuals participate in cross-filtering via Power BI interactions.

4.1 Visuals

Visual	Type	Insight
Profit / Sales / Margin / Order Count	KPI Cards (4×)	Overall business snapshot
Profit by State	Horizontal bar	Top geographies
Sales by Segment	Pie chart	Revenue mix by customer type
Profit by Category	Column chart	Category comparison
Profit by Month & Year	Line / area chart	Seasonality & trend
Sales vs Profit by SubCategory	Scatter plot	Quadrant analysis

4.2 DAX Measures

Measure	DAX Expression	Purpose
Total Sales	SUM(Orders[Sales])	Revenue baseline
Total Profit	SUM(Orders[Profit])	Profit baseline
Profit Margin %	DIVIDE([Total Profit], [Total Sales], 0)	Efficiency ratio
AOV	DIVIDE([Total Sales], DISTINCTCOUNT(Orders[OrderID]), 0)	Average order value
Sales Last Year	CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Calendar'[Date]))	YoY comparison

5. Key Insights

5.1 Profitability

- New York and California are the most profitable states, contributing the largest margin share.
- Technology generates the highest profit per unit despite lower volume than Office Supplies.
- Furniture (Tables, Bookcases) consistently produces negative profit — aggressive discounting is the root cause.

5.2 Customer Segments

- Consumer segment drives 54.95% of total sales; Corporate contributes 26.52%; Home Office 18.53%.
- Corporate orders carry higher average order values despite lower revenue share.

5.3 Seasonality

- Profit dips in month 6 (June) across all years; Q4 (months 10–12) is consistently the peak period.
- Year-over-year revenue and profit growth is positive, with 2017 outperforming all prior years.

5.4 Recommendations

- Restrict deep discounts on Furniture sub-categories to eliminate loss-making orders.
- Invest in Technology marketing ahead of Q4 to capitalise on the highest-margin period.
- Target Corporate segment with bundled offers to grow the high-AOV customer base.

6. Conclusion

This project successfully demonstrates a complete data analysis pipeline - from relational database normalisation and SQL querying through to interactive Power BI dashboarding and DAX-powered time-intelligence. The decision-driven approach applied at each stage ensures that every technical choice is grounded in a clear business question, producing insights that are immediately actionable for retail management.